

Homework 5

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Problem 1. *This is really a problem in real analysis rather than complex analysis, but the issue that it illustrates – namely, that the mere existence of the partial derivatives $\partial_x u$, $\partial_y u$ at a point (x, y) does not guarantee that u is differentiable at (x, y) – has come up so often in this course that it is worth a problem. Define a function u on $\mathbb{R}^2$ by*

$$u(x, y) = \begin{cases} \frac{xy}{x^2 + y^2}, & (x, y) \neq 0 \\ 0, & (x, y) = 0 \end{cases}.$$

Show that $\partial_x u$ and $\partial_y u$ exist at 0 but that u is not even continuous at 0, let alone differentiable there.

Proof. a ■

Problem 2. *In the proof of Goursat's theorem given in class, we assumed that*

$$\int_{\partial R} f = \alpha \neq 0$$

and got a contradiction. The final step of the proof was that $\alpha/4^n < \alpha/4^n$, which would be a contradiction even if α were 0. So where in the proof was it absolutely essential to assume that $\alpha \neq 0$?

Proof. idfk ■

Problem 3. *Let R be as in ??, and let $f(z) = 1/z$. By direct calculation, show that*

$$\int_{\partial R} f = 2\pi i.$$

Proof. Recall that

$$\frac{1}{z} = \frac{\bar{z}}{|z|^2}.$$

Choose a unit speed parameterization of ∂R such that

$$\begin{aligned} \oint_{\partial R} f dz &= \oint_{\partial R} f(dx + i dy) \\ &= \int_{-1}^1 \frac{x+i}{x^2+1} dx + \int_{-1}^1 \frac{1-iy}{y^2+1} i dy + \int_1^{-1} \frac{x-i}{x^2+1} dx + \int_1^{-1} \frac{-1-iy}{y^2+1} i dy \\ &= \int_{-1}^1 \frac{x+i+i+x-x+i+i-x}{x^2+1} dx \\ &= \int_{-1}^1 \frac{4i}{x^2+1} dx \\ &= 4i (\arctan(1) - \arctan(-1)) \\ &= 4i \left(\frac{\pi}{4} - \frac{-\pi}{4} \right) \\ &= 2\pi i. \end{aligned}$$
■

Problem 4. Let γ be the unit circle with counterclockwise orientation. For any real number λ , give an example of a nonconstant holomorphic function on the annulus $\Delta(2,0) \setminus \Delta(1/2,0)$ such that

$$\frac{1}{2\pi i} \oint_{\gamma} F = \lambda.$$

Proof. Set $F(z) = \lambda/z$ and see problem 3. ■